| dℓ en 0 |
| --- |
| $e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \underset{x \to 0}{o}\left(x^n\right)$ |
| $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n+1}\frac{x^n}{n} + \underset{x \to 0}{o}\left(x^n\right)$ |
| $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \underset{x \to 0}{o}\left(x^{2n}\right)$ |
| $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \underset{x \to 0}{o}\left(x^{2n+1}\right)$ |
| $\operatorname{sh} x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + \underset{x \to 0}{o}\left(x^{2n+1}\right)$ |
| $\operatorname{ch} x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \underset{x \to 0}{o}\left(x^{2n}\right)$ |
| $\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n x^n + \underset{x \to 0}{o}\left(x^n\right)$ |
| $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n} + \underset{x \to 0}{o}\left(x^{2n}\right)$ |
| $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + \underset{x \to 0}{o}\left(x^n\right) \quad (\alpha \in \mathbb{R})$ |
| $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \underset{x \to 0}{o}\left(x^{2n+1}\right)$ |
| $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \underset{x \to 0}{o}\left(x^5\right)$ |